

Bidding Metrics

2024-09-18

- Let n_i^{task} = Number of projects of the same task held by the contractor in the three quarters before starting project i
- Let n_i = Number of projects held by the contractor in the three quarters before starting project i
- Define:
 - Inexperienced contractor $_i$ = 1 if $n_i^{task} = 0$
 - New entrant $_i$ = 1 if $n_i = 0$
 - Level of inexperience $_i$ = $1 - \frac{n_i^{task}}{n_i}$ if $n_i > 0$
 - Level of inexperience $_i$ = 1 if $n_{i,k,t} = 0$.

1 Effect on bids

2 Effect on Delays

2.1 Old vs New Entrant

3 Experienced vs Inexperienced

4 Output

	Experienced			Inexperienced		
	Number of offers I	Initial duration II	Initial budget (000s) III	Number of offers IV	Initial duration V	Initial budget (000s) VI
Treat	1.31*** (0.03)	-28.38*** (1.15)	-168.22*** (8.80)	1.08*** (0.05)	-7.50*** (2.20)	-161.72*** (19.42)
Treat $\times$ SA $_i$	0.10** (0.04)	4.57*** (1.35)	-21.27** (10.81)	-0.02 (0.09)	-11.99*** (3.31)	-34.23 (29.94)
Task FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	110,222	108,508	110,258	37,646	36,120	37,702
R ²	0.22	0.27	0.23	0.23	0.29	0.29
Within R ²	0.02	0.01	0.01	0.01	0.00	0.00

Clustered (Project ID) standard-errors in parentheses
 Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

	Percentage delay rate							
	Experienced contractors				Inexperienced contractors			
	I	II	III	IV	V	VI	VII	VIII
Constant	10.965*** (0.197)				12.184*** (0.221)			
Treat	-2.753*** (0.187)	-2.740*** (0.187)	-0.117 (0.193)	-0.181 (0.192)	-1.545*** (0.198)	-1.579*** (0.196)	-1.058*** (0.213)	-1.042*** (0.215)
SA _i	0.763*** (0.224)	1.204*** (0.244)	0.855*** (0.238)	0.806*** (0.237)	2.545*** (0.354)	3.970*** (0.370)	4.192*** (0.393)	4.110*** (0.394)
Ln(Stage)	3.049*** (0.047)	3.121*** (0.050)	2.872*** (0.047)	2.839*** (0.047)	4.011*** (0.063)	4.270*** (0.068)	4.246*** (0.069)	4.225*** (0.069)
Post	-1.548*** (0.236)				-1.644*** (0.273)			
Treat × Post	0.586** (0.267)	0.563** (0.268)	0.158 (0.256)	0.146 (0.255)	0.320 (0.310)	0.361 (0.310)	0.538* (0.316)	0.564* (0.316)
Treat × SA _i × Post	0.854*** (0.255)	0.863*** (0.256)	0.820*** (0.248)	0.849*** (0.247)	0.522 (0.407)	0.395 (0.407)	0.084 (0.426)	0.147 (0.428)
Time FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Task FE	No	No	Yes	Yes	No	No	Yes	Yes
Industry FE	No	No	No	Yes	No	No	No	Yes
Observations	113,499	113,499	113,499	113,499	71,383	71,383	71,383	71,383
R ²	0.05	0.05	0.13	0.14	0.07	0.07	0.12	0.13
Within R ²		0.05	0.04	0.04		0.07	0.06	0.06

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***, 0.01, **, 0.05, *, 0.1*

	Old entrants			New entrants		
	Number of offers I	Initial duration II	Initial budget (000s) III	Number of offers IV	Initial duration V	Initial budget (000s) VI
Treat	1.28*** (0.03)	-23.87*** (1.07)	-164.01*** (8.48)	1.02*** (0.08)	-11.78*** (3.53)	-173.81*** (31.93)
Treat × SA _i	0.10*** (0.04)	1.45 (1.29)	-24.45** (10.59)	-0.42*** (0.15)	-10.40* (5.85)	-11.47 (55.69)
Task FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	131,849	129,434	131,905	16,019	15,194	16,055
R ²	0.21	0.27	0.25	0.28	0.32	0.30
Within R ²	0.02	0.01	0.01	0.01	0.00	0.00

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***, 0.01, **, 0.05, *, 0.1*

	Percentage delay rate							
	Old entrants				New entrants			
	I	II	III	IV	V	VI	VII	VIII
Constant	11.396*** (0.177)				11.915*** (0.265)			
Treat	-2.758*** (0.166)	-2.745*** (0.166)	-0.416** (0.171)	-0.438*** (0.170)	-1.019*** (0.235)	-1.114*** (0.233)	-0.808*** (0.264)	-0.852*** (0.266)
SA _i	0.879*** (0.202)	1.399*** (0.220)	1.274*** (0.218)	1.247*** (0.218)	3.542*** (0.565)	5.235*** (0.583)	5.772*** (0.624)	5.680*** (0.628)
Ln(Stage)	3.207*** (0.042)	3.305*** (0.045)	3.117*** (0.044)	3.086*** (0.043)	4.169*** (0.084)	4.467*** (0.091)	4.433*** (0.094)	4.414*** (0.094)
Post	-1.675*** (0.211)				-1.569*** (0.342)			
Treat × Post	0.717*** (0.237)	0.700*** (0.238)	0.257 (0.230)	0.257 (0.229)	0.127 (0.394)	0.255 (0.393)	0.501 (0.406)	0.535 (0.407)
Treat × SA _i × Post	0.816*** (0.231)	0.813*** (0.231)	0.808*** (0.228)	0.816*** (0.227)	0.181 (0.653)	-0.038 (0.653)	-0.624 (0.691)	-0.551 (0.697)
Time FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Task FE	No	No	Yes	Yes	No	No	Yes	Yes
Industry FE	No	No	No	Yes	No	No	No	Yes
Observations	141,434	141,434	141,434	141,434	43,448	43,448	43,448	43,448
R ²	0.06	0.06	0.12	0.13	0.07	0.07	0.13	0.14
Within R ²		0.05	0.05	0.05		0.07	0.06	0.06

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*